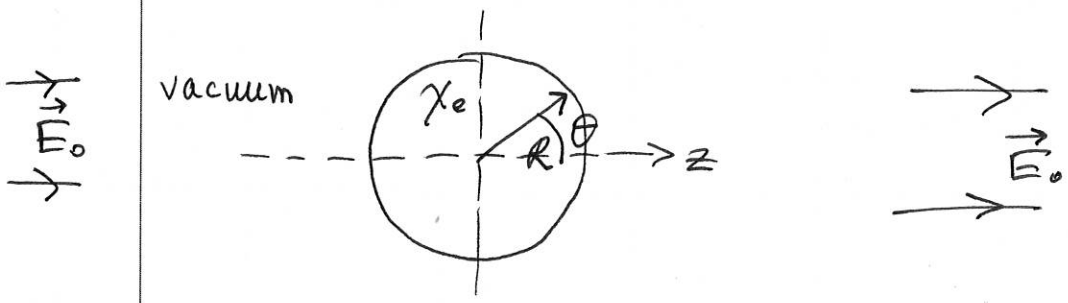
P1 Static situation, no external charges.

$$\vec{\nabla}\times\vec{E}=0$$

$$\vec{\nabla}\cdot\vec{D}=0 \qquad \vec{P}=\varepsilon_0\chi_e\vec{E} \Rightarrow \vec{D}=\varepsilon_0\vec{E}+\vec{P}=\varepsilon_0(1+\chi_e)\vec{E}$$

$$=\varepsilon_0\varepsilon_r\vec{E}=$$

$$=\varepsilon\vec{E}.$$



Hence $\vec{E}=-\vec{\nabla}\phi$ (so that $\vec{\nabla}\times\vec{E}=0$ is satisfied)

and $\vec{\nabla}\cdot\vec{D}=-\vec{\nabla}\cdot(\varepsilon_0\varepsilon_r\vec{E})=-\varepsilon_0\varepsilon_r\nabla^2\phi=0$

in regions with <u>constant $\varepsilon_r=1+\chi_e$</u>

Discussion on pages 295-6 of the textbook shows that in the case of uniformly polarized sphere electric field <u>outside</u> is that of a dipole $\vec{p}$

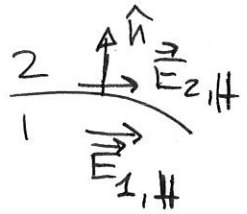
$$\vec{E}_{out}\sim\frac{3\hat{r}(\vec{p}\cdot\hat{r})-\vec{p}}{r^3}$$

while it is uniform inside $\vec{E}_{in}\sim\vec{p}$.

For linear dielectric $\vec{p}\sim\vec{E}_0$ so that we can try the following solution

$$\begin{cases}\vec{E}(\vec{r})=A\vec{E}_0+B\dfrac{3\hat{r}(\vec{E}_0\cdot\hat{r})-\vec{E}_0}{r^3}, & r>R\\ \vec{E}(\vec{r})=C\vec{E}_0, & r<R\end{cases}$$

Boundary conditions:



$\hat{n} = \hat{r}$

a) $\vec{E}_{1\parallel} = \vec{E}_{2\parallel}$

or, $(\vec{E}_2 - \vec{E}_1) \times \hat{n} = 0.$

b) $E_{2,\perp} - E_{1,\perp} =$

$= (\vec{E}_2 - \vec{E}_1) \cdot \hat{n} = \dfrac{\sigma_f + \sigma_b}{\varepsilon_0}$

$\sigma_f = 0$ but $\sigma_b = \hat{P} \cdot \hat{n}$

But for $\vec{D} = \varepsilon_0 \varepsilon_r \vec{E}$ boundary condition is just

$$(\vec{D}_2 - \vec{D}_1) \cdot \hat{n} = \sigma_f = 0$$

$$\vec{D} = \begin{cases} \varepsilon_0 \varepsilon_r \vec{E} & , r < R \\ \varepsilon_0 \vec{E} & , r > R \end{cases}$$

$\Rightarrow$ normal components: $\varepsilon_0 \varepsilon_r \vec{E} \cdot \hat{n} \Big|_{r \leq R} = \varepsilon_0 \vec{E} \cdot \hat{n} \Big|_{r \geq R}$

$$\Rightarrow C \varepsilon_r \vec{E}_0 \cdot \hat{r} = A \vec{E}_0 \cdot \hat{r} + B \frac{3 \hat{r} \cdot \hat{r} (\vec{E}_0 \cdot \hat{r}) - \vec{E}_0 \cdot \hat{r}}{R^3}$$

$$\Rightarrow C \varepsilon_r = A + \frac{2B}{R^3} \quad (1)$$

$\Rightarrow$ tangential components: $\vec{E} \times \hat{n} \Big|_{r \leq R} = \vec{E} \times \hat{n} \Big|_{r \geq R}$

$$C \vec{E}_0 \times \hat{r} = A \vec{E}_0 \times \hat{r} - \frac{B}{R^3} \vec{E}_0 \times \hat{r}$$

$$\Rightarrow C = A - \frac{B}{R^3} \quad (2)$$

Far from the sphere $\vec{E}(\vec{r}) \to \vec{E}_0 \Rightarrow A = 1$.

$\Rightarrow$ (1) + (2) gives: $(\varepsilon_r + 1)\, C = \frac{B}{R^3} + 2$

(1) − (2) gives: $(\varepsilon_r - 1)\, C = \frac{3B}{R^3}$

$$\Rightarrow (\varepsilon_r + 1)\, C = 2 + \frac{1}{3}(\varepsilon_r - 1)\, C$$

$$\left(\frac{2}{3}\varepsilon_r + \frac{4}{3}\right) C = 2 \quad , \quad C = \frac{6}{2\varepsilon_r + 4} = \frac{3}{\varepsilon_r + 2}$$

$$B = \frac{R^3}{3}(\varepsilon_r - 1)\frac{3}{\varepsilon_r + 2} = R^3 \frac{\varepsilon_r - 1}{\varepsilon_r + 2}$$

Or, in terms of $\chi_e = \varepsilon_r - 1$:

$$C = \frac{3}{3 + \chi_e} \quad , \quad \frac{B}{R^3} = \frac{\chi_e}{3 + \chi_e}$$

<u>Solution</u>

$r > R$: $\vec{E}(\vec{r}) = \vec{E}_0 + \frac{\chi_e}{3 + \chi_e} \frac{R^3}{r^3}\left(3\hat{r}\,(\vec{E}_0 \cdot \hat{r}) - \vec{E}_0\right)$

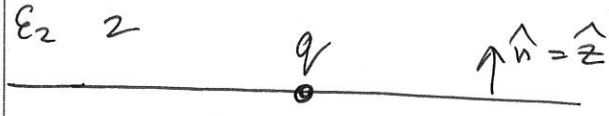
$r < R$: $\vec{E}(\vec{r}) = \frac{3}{3 + \chi_e}\vec{E}_0$

Now can find polarization inside the sphere:

$$\vec{P} = \varepsilon_0 \chi_e \vec{E} = \frac{3\chi_e \varepsilon_0}{3 + \chi_e}\vec{E}_0$$

Total dipole moment: $\vec{p} = \int \vec{P}\, d^3\vec{r} = \frac{4\pi}{3} R^3 \vec{P} = \frac{4\pi\varepsilon_0 \chi_e}{3 + \chi_e}\vec{E}_0 R^3$

P2 let charge q be at the origin.



let $\vec{E} \sim q\frac{\vec{r}}{r^3} = q\frac{\hat{r}}{r^2}$ $\qquad \vec{E} = \begin{cases} A_1 \frac{q\vec{r}}{r^3}, & \text{for region 1 } (z<0) \\ A_2 \frac{q\vec{r}}{r^3}, & \text{for region 2 } (z>0) \end{cases}$

(1) This choice ensures boundary condition for the normal components $(\vec{D}_2 - \vec{D}_1)\cdot\hat{n} = \sigma_f = 0$

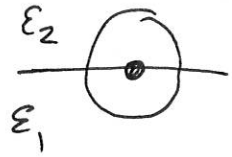
everywhere except for $\vec{r} = 0$ (where the charge is)

(2) Transverse components

$$(\vec{E}_2 - \vec{E}_1)\times\hat{n} = 0 \Rightarrow \frac{q}{r^3}(A_2 - A_1)\,\vec{r}\times\hat{z} = 0$$

$$\Rightarrow A_2 = A_1 = A.$$

Note that consideration in (1) avoids $\vec{r} = 0$ point where the charge is located. In terms of charge density $\rho_f = q\,\delta(\vec{r}) = q\,\delta(x)\,\delta(y)\,\delta(z)$ this is a very singular point.



However $\vec{\nabla}\cdot\vec{D} = \rho_f \Rightarrow \int \vec{\nabla}\cdot\vec{D}\,dV = \int \vec{D}\cdot d\vec{S} = \int \rho_f\,dV =$
$= q$

Because of the assumed spherical symmetry

$$\vec{D} = \vec{D}_1 = A\varepsilon_1 \frac{q\hat{r}}{r^2} \quad \text{for } z<0$$

$$\vec{D} = \vec{D}_2 = \varepsilon_2 A \frac{q\hat{r}}{r^2} \quad \text{for } z>0$$

$$\Rightarrow \int \vec{D}\cdot d\vec{S} = \varepsilon_1 A \frac{q\hat{r}}{r_0^2}\cdot 2\pi r_0^2 \hat{r} + \varepsilon_2 A \frac{q\hat{r}}{r_0^2}\cdot 2\pi r_0^2 \hat{r} = q$$

choose gaussian surface to be a sphere of radius r_0 $\Rightarrow$ area of a <u>hemisphere $= 2\pi r_0^2$</u>

$$\Rightarrow 2\pi A q(\varepsilon_1+\varepsilon_2) = q \Rightarrow A = \frac{1}{2\pi(\varepsilon_1+\varepsilon_2)}$$

$$\Rightarrow \vec{D} = \begin{cases} \dfrac{\varepsilon_1 q}{2\pi(\varepsilon_1+\varepsilon_2)} \dfrac{\hat{r}}{r^2} & z<0 \\[2ex] \dfrac{\varepsilon_2 q}{2\pi(\varepsilon_1+\varepsilon_2)} \dfrac{\hat{r}}{r^2} & z>0 \end{cases}$$

$$\Rightarrow \vec{E} = \frac{q}{2\pi(\varepsilon_1+\varepsilon_2)} \frac{\hat{r}}{r^2} \quad \text{for all } \vec{r}.$$

$$\vec{E} = -\vec{\nabla}\varphi \Rightarrow \varphi(\vec{r}) = \frac{q}{2\pi(\varepsilon_1+\varepsilon_2) r}$$

P2

$$\varepsilon(\omega) = \varepsilon_0\left(1+\chi_e(\omega)\right), \quad \varepsilon'' = \varepsilon_0\,\chi_e'' = \frac{\beta_0\omega\tau}{1+\omega^2\tau^2}.$$

Kramers-Kronig $\quad \chi'(\omega) = \frac{2}{\pi} P\int_0^\infty \frac{x\,\chi''(x)}{x^2-\omega^2}\,dx$

$$\chi'(\omega) = \frac{2}{\pi}\,\frac{\beta_0}{\varepsilon_0}\,P\int_0^\infty dx\; x\,\frac{x\tau}{1+x^2\tau^2}\,\frac{1}{x^2-\omega^2} = \begin{pmatrix} y = x\tau \\ dx = \frac{dy}{\tau}\end{pmatrix}$$

$$= \frac{2\beta_0}{\pi\varepsilon_0}\,P\int_0^\infty \frac{dy}{\tau}\,\frac{y^2}{\tau(1+y^2)}\,\frac{\tau^2}{y^2-\omega^2\tau^2} =$$

$$= \frac{2\beta_0}{\pi\varepsilon_0}\,P\int_0^\infty dy\,\frac{y^2}{1+y^2}\,\frac{1}{y^2-\omega^2\tau^2} = \left(\text{denote } \omega^2\tau^2 = u^2\right)$$

$$= \frac{2\beta_0}{\pi\varepsilon_0}\,P\int_0^\infty dy\left(\frac{1}{1+y^2}+\frac{u^2}{y^2-u^2}\right)\frac{1}{1+u^2}$$

$$\frac{y^2}{(1+y^2)(y^2-u^2)} = \left(\frac{1}{1+y^2}+\frac{u^2}{y^2-u^2}\right)\frac{1}{1+u^2}.$$

$$I_1 = P\int_0^\infty dy\,\frac{1}{1+y^2} = \frac{\pi}{2}$$ (Principal part "P" does not play a role since the integral is regular)

$$I_2 = P\int_0^\infty \frac{dy}{y^2-u^2} = (y=ux) = \frac{1}{u}\,P\int_0^\infty dx\,\frac{1}{x^2-1} = \frac{1}{u}\int_0^\infty dx\,\frac{1}{2}\left(\frac{1}{x-1}-\frac{1}{x+1}\right)$$

$$= \frac{1}{2u}\left\{\int_0^{1-\varepsilon}\frac{dx}{x-1} + \int_{1+\varepsilon}^{L}\frac{dx}{x-1} - \int_0^{L}\frac{dx}{x+1}\right\}.$$

At the end we need to take limits: $\varepsilon\to+0$, $L\to\infty$

$$\int_0^{1-\varepsilon}\frac{dx}{x-1} = (x=1-z) = \int_1^{\varepsilon}\frac{-dz}{-z} = -\int_\varepsilon^1 d(\ln z) = \ln(\varepsilon)$$

$$\int_{1+\varepsilon}^{L} \frac{dx}{x-1} = (x = 1+z) = \int_{\varepsilon}^{L-1} \frac{dz}{z} = \ln \frac{L-1}{\varepsilon}.$$

$$\int_{0}^{L} \frac{dx}{x+1} = \ln\left(\frac{L+1}{1}\right) = \ln(L+1)$$

$$\Rightarrow I_2 = \frac{1}{2u}\left\{\ln(\varepsilon) + \ln\left(\frac{L-1}{\varepsilon}\right) - \ln(L+1)\right\} = (\varepsilon \text{ drops out!})$$

$$= \frac{1}{2u} \ln\left(\frac{L-1}{L+1}\right) \xrightarrow[L\to\infty]{} 0.$$

Thus $\chi'(\omega) = \frac{2\beta_0}{\pi\varepsilon_0} \frac{\pi}{2} \frac{1}{1+(\omega\tau)^2} = \frac{\beta_0}{\varepsilon_0(1+\omega^2\tau^2)}$

$$\chi(\omega) = \chi' + i\chi'' = \frac{\beta_0}{\varepsilon_0(1+\omega^2\tau^2)}(1+i\omega\tau) =$$

$$= \frac{\beta_0(1+i\omega\tau)}{\varepsilon_0(1+i\omega\tau)(1-i\omega\tau)} = \frac{\beta_0}{\varepsilon_0(1-i\omega\tau)}$$

$$\Rightarrow \varepsilon(\omega) = \varepsilon_0 \chi(\omega) = \frac{\beta_0}{1-i\omega\tau}.$$